SOFTWARE TESTING AND DEPLOYMENT

Application Development and Emerging Technologies

WHAT IS SOFTWARE TESTING AND DEPLOYMENT?

- Software Testing: Process of evaluating software to ensure it meets requirements and is free of defects.

- Deployment: The process of releasing the tested software to production or end users.

Goal: Deliver a reliable, secure, and efficient product.

IMPORTANCE OF SOFTWARE TESTING

- Detects bugs before users do 🐛
- Ensures software quality and performance
- Prevents costly post-release failures
- Builds customer trust and satisfaction
- Validates that the system meets business and technical needs

TYPES OF SOFTWARE TESTING

- **Manual Testing** – Performed by QA testers manually.
- **Automated Testing** – Uses tools and scripts for faster, repeatable tests.

LEVELS OF TESTING

Level	Purpose	Example
Unit Testing	Test individual components	Testing a function or method
Integration Testing	Test combined modules	Checking data flow between APIs
System Testing	Test the entire system	Ensuring full workflow correctness
Acceptance Testing	Validate with user requirements	UAT before client sign-off

COMMON TESTING TYPES

Functional Testing – Verifies features work as intended
Non-Functional Testing – Checks performance, security, usability
Regression Testing – Ensures new code doesn’t break existing features
Smoke Testing – Initial check to see if the build is stable
Sanity Testing – Quick test after bug fixes
Performance Testing – Stress, load, and scalability tests
Security Testing – Identifies vulnerabilities and risks

SOFTWARE TESTING PRINCIPLES

1. Testing shows the presence of defects (not their absence)
2. Exhaustive testing is impossible – focus on risk-based testing
3. Early testing saves time and cost
4. Defects cluster together – most errors occur in a few modules
5. Pesticide paradox – change test cases regularly
6. Testing is context-dependent
7. Absence-of-errors fallacy – finding no bugs ≠ good software

Example:

1. Start testing in the requirements phase
2. Focus efforts on high-risk modules
3. Continuously update test cases to adapt to new features or bugs

SOFTWARE TESTING LIFE CYCLE (STLC)

Requirement Analysis
Test Planning
Test Case Design
Environment Setup
Test Execution
Defect Reporting & Retesting
Test Closure

TOOLS USED IN TESTING

Purpose	Tools
Test Management	Jira, TestRail
Automated Testing	Selenium, Cypress, Playwright
Performance Testing	JMeter, LoadRunner
API Testing	Postman, RestAssured
CI/CD Integration	Jenkins, GitHub Actions

WHAT IS DEPLOYMENT?

Deployment is the process of releasing software to the production environment where users can access it.

DEPLOYMENT STAGES

- **Build Preparation –** Compile code, package artifacts
- **Pre-Deployment Testing –** Final sanity & smoke checks
- **Release Deployment –** Move build to staging/production
- **Post-Deployment Verification –** Confirm everything runs as expected
- **Monitoring & Maintenance –** Check logs, fix hot issues

DEPLOYMENT MODELS

Model	Description
Big Bang Deployment	All components released at once
Phased Deployment	Released in stages or modules
Blue-Green Deployment	Two environments alternate between live and test
Canary Deployment	Release to a small user group first
Rolling Deployment	Gradually replace old version with new

TOOLS FOR DEPLOYMENT

- **CI/CD Tools:** Jenkins, GitLab CI, GitHub Actions
- **Containerization:** Docker, Kubernetes
- **Cloud Deployment:** AWS, Azure, GCP
- **Version Control:** Git, GitHub, Bitbucket

THANK YOU